

DUNMAN SECONDARY SCHOOL

CANDIDATE
NAME

CLASS

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NUMBER

PRELIMINARY EXAMINATION 2024 SECONDARY 4 NORMAL ACADEMIC

MATHEMATICS SYLLABUS A

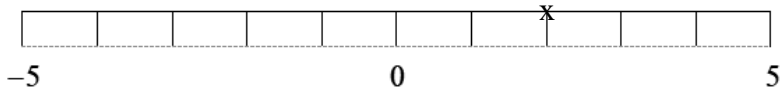
Paper 2

4045/02

5 August 2024

2 hours

Marking Scheme

Question	Answer	Marks
1a	14.27	1
1b		1
1c	724g 0.38kg 9248mg	2

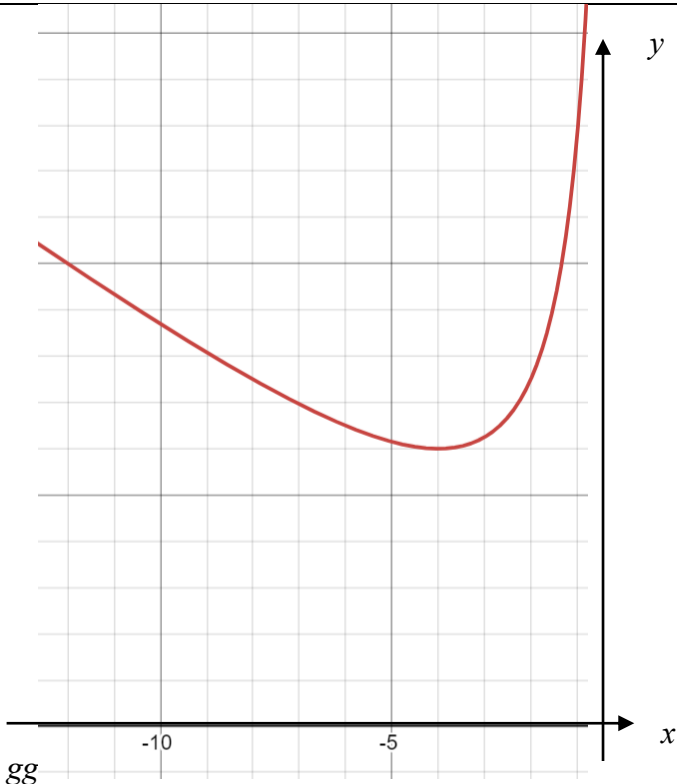
Question	Answer	Marks
2a	$\frac{3 \times 3^{-6}}{(3^2)^2} = \frac{3^{-5}}{3^4}$ $= 3^{-5-4}$ $= 3^{-9}$	2
2b	$\left(\frac{125}{a^9}\right)^{-\frac{2}{3}} = \left(\frac{a^9}{125}\right)^{\frac{2}{3}}$ $= \left(\frac{a^3}{5}\right)^2$ $= \frac{a^6}{25} \text{ OR } 0.04a^6$	2

Question	Answer	Marks
3a	$2.4 = \frac{0(3)+1(5)+2(12)+3(p)+4(6)+5(2)}{3+5+12+p+6+2}$ $2.4(28+p) = 63+3p$ $67.2+2.4p = 63+3p$ $67.2-63 = 3p-2.4p$ $4.2 = 0.6p$ $p = 7$	3
3b	$\frac{8}{(their\ 35)}$	1

Question	Answer	Marks
4a	$\frac{(5-2) \times 180^\circ}{5} = 108^\circ$	1
4b	$\tan 54^\circ = \frac{h}{4}$ $h = 4 \tan 54^\circ$ $h = 5.5055$ Area of one triangle = $\frac{1}{2}(8)(\text{their } 5.5055)$ $= 22.022$ Area of pentagon = 5×22.022 $= 110.11$ $= 110 \text{ (3 s.f)}$	3
4c	Volume = Base area $\times$ Height $3528.5 = (\text{their } 110.11) \times \text{Height}$ $\text{Height} = \frac{3528.5}{(\text{their } 110.11)}$ $= 32.045$ $= 32.0 \text{ (3 s.f)}$	2

Question	Answer	Marks
5a (i)	$180^\circ - 129^\circ = 51^\circ$	1
5a (ii)	$180^\circ - 51^\circ - 33^\circ = 96^\circ$	1
5a (iii)	33°	1
5b	$\angle AED = 180^\circ - 129^\circ$ $= 51^\circ$ $\angle ADE = 180^\circ - 78^\circ - 51^\circ$ $= 51^\circ$ Since angle ADE is equal to angle AED, triangle ADE is isosceles	2

Question	Answer	Marks
6 (a)	$axy = x - b$ $axy - x = -b$ $x(ay - 1) = -b$ $x = \frac{-b}{ay - 1}$	3
6 (b)	$y^2(y + 7) - 16(y + 7) = (y^2 - 16)(y + 7)$ $= (y - 4)(y + 4)(y + 7)$	2
6 (c)	Cost of phone case = \$(x + 4.50) $\$[2x + 5((\text{their } x + 4.50))] = \75 $2x + 5x + 22.50 = 75$ $7x = 75 - 22.50$ $7x = 52.50$ $x = 7.50$ Cost of phone case = \$(7.50 + 4.50) $= \$12$	3
	<u>Alternative Solution</u> cost of 7 casing = \$75 + 2(\$4.50) $= \$84$ cost of 1 casing = \$84 ÷ 7 $= \$12$	3

Question	Answer	
7a	10	
7b		
7c	Drawing tangent that is horizontal $x = -4$	2
7d	$\frac{-3x}{4} - \frac{12}{x} = 7$ $y = 7$ $x = -2.3 \text{ to } -2.1 \text{ and } -7.2 \text{ to } -7$	3

Question	Answer	Marks
8a	$14^2 = AE^2 + 12^2$ $AE^2 = 14^2 - 12^2$ $AE = 7.2111$ $= 7.21 \text{ cm (3 s.f)}$	2
8b	$\tan 35^\circ = \frac{FC}{3}$ $FC = 3 \tan 35^\circ$ $FC = 2.10062$ Area of unshaded $= \frac{1}{2}(12)(\text{their } 7.2111) + \frac{1}{2}(3)(\text{their } 2.10062)$ $= 46.41753$ Area of rectangle $= 12(7.9 + \text{their } 2.10062)$ $= 120.00744$ Area of shaded $= 120.00744 - 46.41753$ $= 73.58991$ $= 73.6 \text{ cm}^2 \text{ (3 s.f)}$	4

Question	Answer	Marks
9a	(8 , 2)	1
9b (i)	$y = 1$ Accept $y = 0x + 1$ (But for national exam they might not accept hence need to simplify to $y = 1$)	1
9b (ii)	$\text{Gradient} = \frac{5-1}{8-3}$ $= \frac{4}{5}$ $y = \frac{4}{5}x + c$ Sub in (3,1) $1 = \frac{4}{5}(3) + c$ $c = 1 - \frac{12}{5}$ $c = -\frac{7}{5}$ BD equation is $y = \frac{4}{5}x - \frac{7}{5}$	3
9c	$\text{Area} = \frac{1}{2}(\text{their } 8)(4) - \frac{1}{2}(\text{their } 5)(1)$ $= 16 - 2.5$ $= 13.5 \text{ unit}^2$	2
9d	$\sqrt{(8-0)^2 + (5-1)^2} = 8.94$	2

Question	Answer	Marks
10a	$\frac{1199}{1199 + 781 + 708 + 379 + 323 + 240 + 123 + 70 + 744} \times 100\% = \frac{1199}{4567} \times 100\%$ $= 26.3\%$	2
10b	Total cost in korean won = 17810000 + 17673000 $= 35483000$ Total cost in SGD = 35483000 ÷ 1024.15 $= S\$34646.29 \text{ (2 d.p)}$	2
10c	<u>Sweden University</u> Total cost in Sweden Krona = 182200 + 129000 $= 311200$ Total cost in SGD = 311200 × 0.1247 $= 38806.64$ <u>Income</u> Annual income = Monthly salary × 12 + Performance Bonus $= 6950 \times 12 + 6950 \times 1.25$ $= \text{SGD } 92087.50$ Annual Household Expenditure = (their 4567) × 12 $= \text{SGD } 54804$ Amount left for overseas study = (their 92087.50) – (their 54804) $= \text{SGD } 37283.50$ Mr Mah is unable to afford to send his child to Sweden.	4

Question	Answer	Marks
11a(i)	$\text{Mean} = \frac{5(9) + 15(28) + 25(41) + 35(53) + 45(34) + 55(15)}{180}$ $= 31.7 \text{ (3 s.f)}$ $\text{S.D} = 13.1$	3
11a(ii)	School <i>A</i> has a wider variation as their standard deviation of 13.1 is more than School <i>B</i> 's 12.9.	2
11b	$P(\text{diff colour}) = \left(\frac{4}{7} \times \frac{3}{7}\right) + \left(\frac{3}{7} \times \frac{4}{6}\right)$ $= \frac{26}{49}$	3

Question	Answer	Marks
12a(i)	$\angle SRP = 103^\circ$ (angles in same segment)	1
12a(ii)	$\text{Reflex } \angle SOP = 103^\circ \times 2$ (Angle at centre = $2 \times$ angle at circumference) $= 206^\circ$ $\angle SOP = 360^\circ - 206^\circ$ (Angles at a point) $= 154^\circ$	2
12b(i)	$\frac{AB}{\sin 97.3^\circ} = \frac{7.5}{\sin 36.2^\circ}$ $AB = \frac{7.5 \sin 97.3^\circ}{\sin 36.2^\circ}$ $AB = 12.5958$ $AB = 12.6 \text{ (3 s.f)}$	1
12b (ii)	$9.3^2 = 7.5^2 + 9^2 - 2(7.5)(9)\cos \angle DBC$ $\frac{9.3^2 - 7.5^2 - 9^2}{-2(7.5)(9)} = \cos \angle DBC$ $\angle DBC = \cos^{-1} \left[\frac{9.3^2 - 7.5^2 - 9^2}{-2(7.5)(9)} \right]$ $\angle DBC = 67.9138^\circ$ $= 67.9^\circ$	2
12b (iii)	$270^\circ + (\text{their } 67.9^\circ) = 337.9^\circ$ (1 d.p)	1
	<u>Alt Solution</u> $90^\circ - (\text{their } 67.9^\circ) = 22.1^\circ$ $360^\circ - (\text{their } 22.1^\circ) = 337.9$	

–1m from entire question if no degree symbol written at all
–1m from entire question if no reason stated